Exp.no : 19 **NUMPY LIBRARY**

AIM:

Understand array function using Numpy library.

PYTHON CODE:

```
import numpy as np

# Create an array
arr = np.array([10, 20, 30, 40, 50])

print("Original Array:", arr)

# Array dimensions
print("Dimensions:", arr.ndim)

# Array shape
print("Shape:", arr.shape)

# Array size
print("Size:", arr.size)

# Sum of array elements
print("Sum:", np.sum(arr))

# Mean of array elements
print("Mean:", np.mean(arr))

# Maximum and minimum values
print("Maximum:", np.max(arr))
print("Minimum:", np.min(arr))

# Sort an array
arr2 = np.array([50, 10, 40, 20, 30])
print("Sorted Array:", np.sort(arr2))

# Reshape an array
arr3 = np.arange(1, 7).reshape(2, 3)
print("Reshaped Array:\n", arr3)

# Array arithmetic
print("Addition:", arr + 5)
```

```
print("Multiplication:", arr * 2)
```

OUTPUT:

```
Original Array: [10 20 30 40 50]
Dimensions: 1
Shape: (5,)
Size: 5
Sum: 150
Mean: 30.0
Maximum: 50
Minimum: 10
Sorted Array: [10 20 30 40 50]
Reshaped Array:
 [[1 2 3]
  [4 5 6]]
Addition: [15 25 35 45 55]
Multiplication: [ 20  40  60  80 100]
```

RESULT:

The code is executed successfully.